

Focused review of the aligned paired circuit

Andrew Barnes October 7, 2026

Status: review requested; no external assessment recorded. This packet builds on Douglas Colkitt's paired construction and OpenAI's *Integer multiplication below $n \log n$* . The complete upstream theorem remains assumed. Pinned manuscript commit:

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

Original aligned contribution in PR #2:

a961c43eecacb72929021936798dc049771f8536.

Finite claim F. For formal inputs x_S indexed by three-element subsets of $\{0, \dots, 49\}$, every three-element target T and $i \in T$ has

$$y_{i,T} = \sum_{S \cap T = \{i\}} x_S.$$

The graph uses 435,450 binary additions with disjoint supports and 58,800 partial outputs: 14,944 fewer additions than the preceding paired graph. Summing the three outputs for T gives the intersection-one map. These finite identities and counts are independently reproducible.

Conditional consequence C. Assume the retained embedding and motif argument realize the stated finite shear contract, the power-width and arbitrary-width interchange procedures have their claimed uniform fixed-tape costs, and the retained layer, stopped-guard, resampling, transform, precision and assembly interfaces apply. Then

$$T(n) = O(n(\log n)^{1-17 \cdot 2^{-63}}).$$

The machine has a fixed finite alphabet and a fixed finite number of one-dimensional tapes. The network is fixed independently of n . No worldwide priority or practical speedup is established here.

Construction and local frames. For common point i , order the other points increasingly except for $i \oplus 1$, which goes last. Relabel inputs and outputs of the unchanged weighted paired circuit (9,813 additions, 1,176 outputs). Complete root pairs agree across groups. Intern equal fixed-pair star supports, retain the first decomposition, prune and recompile physical roles.

For node support A , let U_A span its source indicators in the rational form $B = I - J/9$ on $\mathbb{Q}^{50}$. The sources share a point i , so

$$\sum_j u_j = 3u_i, \quad u^T B u = \sum_{j \neq i} u_j^2 \quad (u \in U_A).$$

Zero forces $u = 0$. Thus U_A is positive definite, and $U_A^\perp$ is nondegenerate because the ambient form is nondegenerate ($50 \neq 9$). At an output $1_S^T B 1_T = |S \cap T| - 1 = 0$. Support inclusion nests forward spans; orthogonal complements reverse inclusions for the backward graph. These local premises do not by themselves prove the entire tensor/motif transfer.

Gather/fanout compilation has $2C + Q$ outgoing uses and reuses one incoming role per addition, giving $R = C + Q = 494250$ side roles. The retained schedule cancels JLz and $JL(z + Vx)$ for arbitrary initial scratch z , leaving $JLVx$ and restoring scratch. Here L, V, J denote mixers, source copy and output injection. **R1:** check the full forward/reverse schedule, tensor frames, first/third-stage matching and every role's endpoint. The negative X-source correction adds N ranks. Asserted counts:

$$\begin{aligned} h &= 50, & v &= 19600, & N &= v^3, & m &= 125000, & L_{\text{loss}} &= 3v^2h^2, \\ W &= 2N + 2v^2(R + h) = 394839648000000, \\ s &= Wm - N + 2L_{\text{loss}} = 49354954232864000000, \\ D &= Wm - s = 1767136000000, & \eta &= D/(Wm) = 23/642375000. \end{aligned}$$

R2: circuit to fixed-tape recurrence

The required contract routes every scalar input by a permutation ρ of all W roles, including arbitrary scratch. Each gate has one common rational $m \times m$ matrix M ; its edge matrices satisfy

$$M_{\text{out}(\rho(w))} - M_{\text{in}(w)} = I_m \quad (1 \leq w \leq W),$$

$$s = \sum_e \text{rank}_{\mathbb{Q}}(M_{\text{head}(e)} - M_{\text{tail}(e)}) < Wm.$$

This is `eq:finite-shear-contract` in `04-swap.tex`. A correct sum circuit alone does not imply the contract.

Logical volume. In `prop:power-interchange`, width $e = m^k$ has a preceding row count divisible by W^k . Cyclic splitting creates W role streams of the same remaining shape, each with logical volume $V_{\log}/W$ and row count divisible by W^{k-1} . By `lem:matrix-shear`, each edge shear uses exactly its rational rank in recursive pivot interchanges. Every spectator range remains present. One fixed prime avoids the finite rational table's denominators and invertible-factor diagonal numerators.

Uniform node bound to audit. There must be exactly s child calls on volume $V_{\log}/W$, with at most $C_0 V_{\log}$ other tape steps. C_0 may depend on the fixed network and prime, but not on depth, field lengths or data. The existing proof reserves W role tapes, fixed I/O/work tapes, and parking and descriptor stacks. It parks inactive streams, copies the active stream to child I/O, erases current parent work and pushes its descriptor. On return, streams are restored from the stack top without traversing ancestors. Charge every copy, erasure and head movement. A fixed tape count can be huge.

Descriptors and one-bit payloads. The pinned proof includes both a parking schedule and a descriptor argument. At depth j of root width $e_0 = m^{k_0}$, all original chunk ranges remain as targets or spectators:

$$V_j = V_0/W^j \geq q^{2e_0}, \quad j \log W \leq (\log_m e_0) \log W = O(e_0), \quad \log V_0 = O(\log(2V_j)).$$

The child keeps its two targets and collapses consecutive spectators into a bounded number of intervals. Its descriptor processing has fixed polynomial cost in $\log(2V_0)$, absorbable in $O(V_j)$. Check every allowed shape and that cleanup stays within the current call's extent. The question is the validity of this existing argument, not the absence of a schedule.

Conditional recurrence. Under that uniform node bound, time divided by logical volume satisfies

$$F_0 = O(1), \quad F_k \leq (s/W)F_{k-1} + C_0.$$

The exact enclosure gives $\log m < 11737/1000$ and $\eta > a(11737/1000)$ with $a = 305/10^{11}$. Therefore

$$1 - \eta < \exp(-a \log m), \quad s/W < m^{1-a},$$

and the geometric recurrence gives $O(V_{\log} e^{1-a})$. `lem:chunk-swap` must preserve this exponent when removing row/width restrictions, including padding and cleanup. A fixed number s of parking operations fits in C_0 ; using total workspace as the child's logical volume loses the claimed coefficient s/W .

Requested assessment. An expert uninvolved in the original construction or audit should examine this uniform bound, complete address ranges, spectator compression, stack discipline and primitive stream costs. Finite coefficient and dirty-scratch tests do not prove the machine time bound.

R3: downstream implication and verification

Retain $a_c = 1/10^{11}$, $\beta = 999/1000$, $\epsilon = 199/1000$, $\delta = 1/10000$, $C_1 = 2$. Set

$$\tau = 1 - a, \quad \sigma = 1 - a_c, \quad c = \beta a, \quad \lambda = 1 - (1 + \beta)a^2/2, \quad \lambda' = 1 - \beta a^2.$$

The stopped guard uses the unchanged complex motif. The Gaussian width is $\alpha = \lceil (32db)^{1/4} \rceil$, with cutoff $b \geq 2^{40}$. The exact certificate checks 30 strict inequalities and all seven margins:

$$G = \epsilon\beta a^2 = \frac{739738521}{4000000000000000000000000} > 17 \cdot 2^{-63}.$$

R3: verify that these inequalities match actual retained algorithmic hypotheses and costs. Strict rational margins absorb fixed logarithmic factors after the cost, transform, precision and rounding contracts are justified.

Dependency map. LaTeX anchors are in the unchanged pinned source under `upstream/build/sections/`; the aligned patch supplies revised parameters.

- **F:** `scripts/aligned_paired_network.py`, `scripts/shared_point_circuit.py`; complete coefficient oracle in `tests/test_aligned_paired_network.py`.
- **R1:** `03-motifs.tex`: `eq:motif-schedule`, `eq:common-frame-identity`, `prop:bit-motif-interface`; shared-computation and complement refinement proofs in `docs/research/`.
- **R2:** `04-swap.tex`: `lem:matrix-shear`, `eq:finite-shear-contract`, `prop:power-interchange`, `eq:swap-recurrence`, `lem:chunk-swap`; `02-streams.tex`: `lem:elementary-streams`, `lem:ordered-affine-streams`.
- **R3:** `05-layers.tex`: `prop:simultaneous-layer`; paired stopped-guard/Gaussian audit; `07-resampling.tex`: `lem:no-sort-resampling`, `lem:tensor-resampling`; `08-assembly.tex`: `eq:sizes`, `eq:final-error`, `tab:costs`, `eq:margin-list`.

Reproduce the finite evidence. From the repository root, with Python 3 and Git (enter the first command on one line in PowerShell):

```
python -m unittest discover -s tests \
    -p test_aligned_paired_network.py -v
python scripts/aligned_paired_network.py
git diff --exit-code -- certificates/aligned-paired-network.json
git apply --check --directory=upstream patches/h50-aligned-paired.patch
```

Do not use Python's `-O`; several checks use assertions. The six tests cover every full-size DAG coefficient in 1,024-input chunks without `support_in` or compressed provenance; small complete forward/reverse dirty-input basis vectors; both frame directions; and three-stage bank exchange. The certificate separately checks production provenance, symbolic inclusions, counts and exact rational parameters. Run `make verify` for all regressions and historical patch checks. These commands do not prove uniform machine costs.

Response requested. Report separately on R1, R2 and R3: reviewed commit, source anchor, argument or counterexample, and unreviewed prerequisites. Accept a bounded obligation with justification, reject it with a concrete failure/repair, or name an unresolved prerequisite. Finite graph approval alone closes neither R2 nor R3. The complete theorem remains conditional.

Full packet and source links Exact certificate PR #2